BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 240 · Software testing: finding and correcting faults

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Define a cache miss.

Q2. When is A AND B true?

Q3. Give a boundary-invalid input just below 1.

Q4. Does 61 satisfy `seconds>=1 || seconds<=60`?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

logic error — Runs but produces the wrong result.

inclusive boundary — Endpoint allowed by a range.

validation — Check input meets constraints.

debugging — Find and correct a fault.

regression test — Repeat tests after a change.

New learning

Read the stated rule — Because the permitted ages are inclusive, both age 12 and age 18 must be accepted.

Locate the Boolean error — The strict comparison operators > and < exclude both valid boundaries.

Repair and test — Change to >= and <=; retest the boundaries plus values just outside the range.

Worked example

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

1. Predict first — The valid endpoints were incorrectly rejected.

2. Model and check — Change the operators to >= and <=, and test −1 and 11 as invalid values.

We Do · Practise with guidance

A duration validator for 1–60 accepts 61 because it uses OR. Correct it.

```
return seconds >= 1 && seconds <= 60;
```

Predict first

OR accepts 61 because only one side needs to be true.

Model and check

AND requires both limits to hold and correctly rejects 61.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Correct x>5 && x<15 for values 5–15 inclusive, and state whether 5 and 15 are valid.

Question 2

Explain why `age>=12 || age<=18` accepts 100, correct it, and give outputs for 11,12,18,19.

Question 3

Outline a debugging and regression-testing process after correcting a validator.

Exit Ticket · Answer before checking

For a bounded range, OR usually replaces AND without changing the outcome.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Requested result is not already stored.

Q2. Only when both A and B are true.

Q3. 0.

Q4. Yes, because seconds>=1 is true.

You Do pages · Independent questions

Q1. [4] Use the lower inclusive boundary x >= 5 [1] and upper inclusive boundary x <= 15 [1] together with AND: x >= 5 && x <= 15 ; x=5 is accepted [1] and x=15 is accepted [1]. Both clauses must be combined in the final condition.

Q2. [6] 100>=12 is true [1]; use && [1]; 11 false [1], 12 true [1], 18 true [1], 19 false [1].

Q3. [6] Reproduce failing case [1]; identify faulty condition [1]; correct code [1]; retest valid boundaries [1] and invalid neighbours [1]; retain automated regression tests [1].

Exit Ticket · Model answer

False. OR often wrongly accepts values outside the range.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding